Chemistry 3A – Spring 2026 **Version B**

Sections 19320, 19321

Midterm Examination #2

Select the **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

1. Of the following intermolecular forces dipole-dipole (D-D), hydrogen bonding (H-B), London Dispersion Forces (LDF), which choice shows the order of strength of interaction? (LDF > H-B means LDF is stronger than H-B)

   a) LDF > H-B > D-D b) H-B > D-D > LDF c) H-B > LDF > D-D d) D-D > H-B > LDF

2. Which of the following represents a change in enthalpy in a substance that is being is being boiled?

   a) ΔH_{vap} b) ΔH_{sub} c) ΔH_{fus} d) ΔH_{melt}

3. Assume specific heat capacity of cadmium (Cd) is 0.250 J/(g °C). A 10.0 g piece of cadmium metal is heated from 20°C to 50°C. How many joules of heat energy were transferred to the metal?

   a) 400 J b) 225 J c) 75.0 J d) 0.635 J

4. In drawing a Lewis structure for the nitrate $(NO_3)^-$ ion, you have placed the octet around the three oxygen (O) atoms, but you see the central nitrogen (N) atom has only six electrons (3 bonding pairs) to the three oxygen atoms. You have no electrons remaining to add in your inventory. What step do you need to do?

   a) Nothing: your Lewis structure is ready and complete
   b) Take a lone pair from one of the oxygen (O) atoms and use it to create a double bond with the central nitrogen (N)
   c) Add a hydrogen atom to the molecule
   d) Use Avogadro's Number at an earlier step

5. Which of these types of matter will have a definite volume but not a definite shape?

   a) gas b) liquid c) solid d) London Dispersion Force e) choices (a) and (b))

6. What is the molar mass of $CaSO_4$?

   a) 88.14 g/mol b) 104.14 g/mol c) 136.14 g/mol d) 152.14 g/mol

7. What is the best choice for what is TRUE about the characteristics of a Type II ion?

   a) it is a polyatomic anion composed of nonmetal elements
   b) it is an element that will ionize only to one charge state, such as +1 or +2 or +3 or more, but only one of those
   c) the element's name in a compound requires the use of a roman numeral (the Stock system)
   d) it only applies to Group 2 elements, which is why it is called a Type II ion

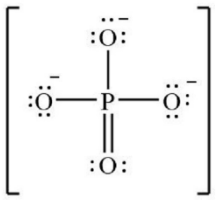
8. On the Periodic Table is a number with each element. For example, for Hydrogen (H), that number is “1.008” or close to that value. What does that number represent?

   a) electronegativity b) grams c) density d) molar mass

9. What is the mass percentage of water in the hydrate $CuSO_4 \cdot 5\ H_2O$?

   a) 7.23% b) 36.08% c) 51.16% d) 63.92%

The Lewis structure is shown for PO_4^{3-} ion at right. **Next four questions** relate to it.



10. What are the total number of electrons (valence electrons of atoms and electrons that account for the charge) for the molecule?

a) 14 b) 18 c) 22 d) 26

11. Each atom in the molecule should have an **octet** of electrons. How many electrons is that?

a) 4 b) 6 c) 8 d) 12

12. Which of the following statements is FALSE about the Lewis structure

a) One of the oxygen atoms has two nonbonding (lone) pairs of electrons
b) There are a total of 3 bonding pairs of electrons in the molecule
c) The molecule has a net charge of -3
d) The central atom phosphate (P) has no nonbonding (lone) pairs of electrons

13. Which of the following is a TRUE statement about the molecule

a) This is an excellent example of a **cation**
b) The central P atom has four (4) **electron groups**
c) There is -1 formal charge on all four O atoms
d) This molecular has a **trigonal planar** molecular geometry

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14. How many moles of a compound is there if the number of molecules or atoms or particles of that compound is five (5) times Avogadro's Number?

a) 0.2 mol b) 1 mol c) 5 mol d) 20 mol

15. Methane (CH_4) is a molecule with four terminal H atoms and no nonbonding (lone) pair of electrons on the central C atom. What is its electron group geometry?

a) bent b) trigonal planar c) trigonal pyramidal d) tetrahedral

16. When atoms like rubidium and chlorine have the largest differences in electronegativity, what kind of bond will they form?

a) ionic b) metallic c) covalent d) polar covalent e) covalent-ionic

17. Water is at 100°C. What is its temperature on the Kelvin scale?

a) 0 K b) 100 K c) 298 K d) 373 K

18. Hydrogen (H) and fluorine (F) show a somewhat large difference in their electronegativity values. But not like sodium (Na) and chlorine (Cl). What kind of chemical bond are H and F likely to form as a molecule?

a) ionic b) metallic c) (nonpolar) covalent d) polar covalent

19. There are 6.022×10^{23} atoms of element potassium (K). How many grams of potassium are there?

a) 6.022×10^{23} g b) 6.65 g c) 40.08 g d) 241.3 g

20. If you have a Ba^{2+} ion and a SO_4^{2-} ion, what compound will it form as you apply the criss-cross method?

a) Ba_2SO_4 b) $Ba_2(SO_4)_2$ c) $BaSO_4$ d) $Ba(SO_4)_2$

21. A diamond is composed of pure carbon (C) and forms an extensive structure of tetrahedral geometry What class of (crystalline) solid does extensively hydrogen-bonded H_2O as ice as well as carbon dioxide (CO_2) as dry ice, and also diatomic element molecule iodine (I_2) form?

a) ionic b) covalent network c) metallic d) molecular

22. If a compound has a molecular formula of $C_4H_{10}N_2$, what would be the empirical formula for it?

a) $C_4H_{10}N_2$ b) $C_8H_{20}N_4$ c) C_2H_5N d) CHN

23. What kind of **intermolecular force** involves a hydrogen atom bonded to a very electronegative atom and causing a strong attraction to a lone (nonbonding) pair of another electronegative atom?

a) London dispersion forces b) hydrogen bonding
c) covalent-ionic attraction d) dipole-dipole interaction

24. What is the correct name for **$FeCl_3$**, noting that Fe is a Type II metal cation?

a) tin(IV) chloride b) iron(II) chloride c) iron(III) chloride d) iron chlorine

25. Which molecular formula correctly shows the compound **dinitrogen tetroxide**?

a) NO b) N_4O_2 c) N_2O_2 d) N_2O_4

26. There are more molecules in 0.50 mol of CO_2 than in 0.50 mol of SO_2
    a) true b) false
27. 1 mol of sodium (Na) and 1 mole of chlorine (Cl) are equal numbers of Na and K atoms
    b) true b) false
28. Calcium is a Group 13 element
    a) true b) false
29. Chemical potential energy is stored energy in a chemical bonds
    a) true b) false
30. Nonbonding (lone) pairs of electrons affecting molecular geometry is explained by Valence Shell Electron Pair Repulsion (VSEPR) theory
    a) true b) false
31. London dispersion forces are the weakest of intermolecular forces where dipole-dipole interactions and hydrogen bonding are the other forces.
    a) true b) false
32. The freezing of water to ice is an endothermic process
    a) true b) false
33. Beryllium has a molar mass of 9.012 g/mol
    a) true b) false
34. Condensation is the process of a solid turning directly into a gas.
    a) true b) false
35. Sulfur dioxide (SO_2) is an ionic compound.
    a) true b) false